0. **Code:** dahiyav6175 Dahiya ISAM 5430 Test 2.zip

1. Dahiya ISAM 5430 Test 2/Test 2 Larger Running Average/Test 2 Larger Running Average/Program.cs
2. Dahiya ISAM 5430 Test 2/Test 2 Nested Loops/Test 2 Nested Loops/Program.cs

1. est 2/Test 2 Larger Running Average/Test 2 Larger Running Average/Program.cs ▶

```
namespace Test_2_Larger_Running_Average
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] arr = {
                7,
                3,
                1,
                8,
                2,
                4,
                5,
                0,
                9
            };
            double newArray = vd(arr);
            Console.Write(string.Join(",", newArray));
        }
        public static double vd(int[] array)
        {
            double sum = 0;
            int prev = 0;
            int count = 0;
            double average = 0;
            int c = 0;
            if (array.Length == null && array.Length == 0)
            {
                Console.WriteLine("There are no valid Values in an array");
            }
```

A1

A3

array = null

return 0

1. est 2/Test 2 Larger Running Average/Test 2 Larger Running Average/Program.cs

```
            for (int i = 0; i < array.Length; i++)

            {
                if (i == 0)
                {
                    prev += array[i];

                }
                if (c ≤ i - 1)
                {
                    prev += array[i];
                    average = (prev / i);

                }
                if (array[i] ≥ average && i ≠ 0)
                {
                    sum += array[i];
                    count++;

                }

            }

            return sum / count;
        }
    }
}
```

A2c

Should be before here because You are calculating the previous value. Same code.

A2b

else?

what's c?

c is always Ø here!

A2

if (count > 0)

need to check Ø!

else return Ø;

A3

2. Dahiya ISAM 5430 Test 2/Test 2 Nested Loops/Test 2 Nested Loops/Program.cs

```
namespace Test_2_Nested_Loops
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] arr = {
                7,
                3,
                1,
                8,
                2,
                4,
                5,
                0,
                9
            };
            int[] newArray = vd(arr);
            Console.Write(string.Join(",", newArray));
        }

        public static int[] vd(int[] array)
        {
            if (array.Length == null && array.Length == 0)
            {
                Console.WriteLine("There are no valid Values in an array");
            }

            int sum = 0;
            for (int i = 0; i < array.Length; i++)
            {
```

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2. Dahiya ISAM 5430 Test 2/Test 2 Nested Loops/Test 2 Nested Loops/Program.cs

```
            sum += array[i];
        }
        int[] newArray = new int[(array.Length * (array.Length + 1))
          / 2];
        int index = 0;

        for (int i = 0; i < array.Length; i++)
        {

            if (array[i] ≥ 0)
            {
                for (int j = 0; j < array.Length; j++)
                {
                    newArray[j] = array[j];
                    index++;
                }
                int min = int.MaxValue;
                for (int k = 0; k < array.Length; k++)
                {
                    if ((array[k] ≤ min))
                    {
                        min = array[k];
                    }
                }
                int start = 0;
                int end = array.Length;
                for (int j = end; j < array.Length; j++)
                {
                    if (newArray[k] > min)
                    {
                        newArray[k] = array[k];
```

why finding the sum?

B2

B4

positive? } - not what? a requirement

outer loop

B2

j is the index for array[j] not newarray. $O(n^2)$

int minIndex = -1

B4

check newarray.

minIndex = k;

check incorrectly.

outside the loop

B3

j is not used

What?

not going into the loop

Index++ j ? no logic here.

2. **Dahiya ISAM 5430 Test 2/Test 2 Nested Loops/Test 2 Nested Loops/Program.cs**

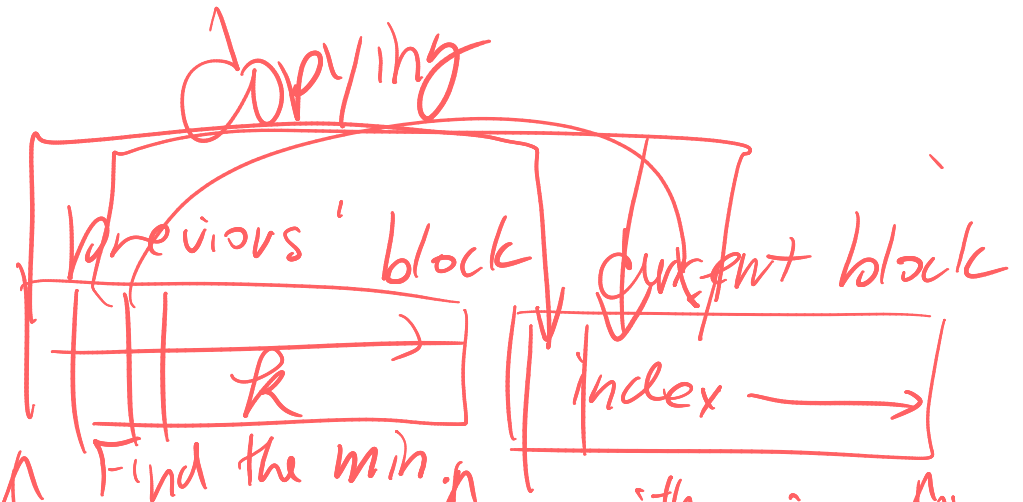
```
                        }
                    }
                }
            }
        }
        return newArray;
    }

    }
}
```

Incomplete

B5 Nested loop - Yes.
Using the correct index - no.

Copying



Find the min.
$(i-1)^{th}$ iteration.

i^{th} iteration

need extra index tracking for offsetting